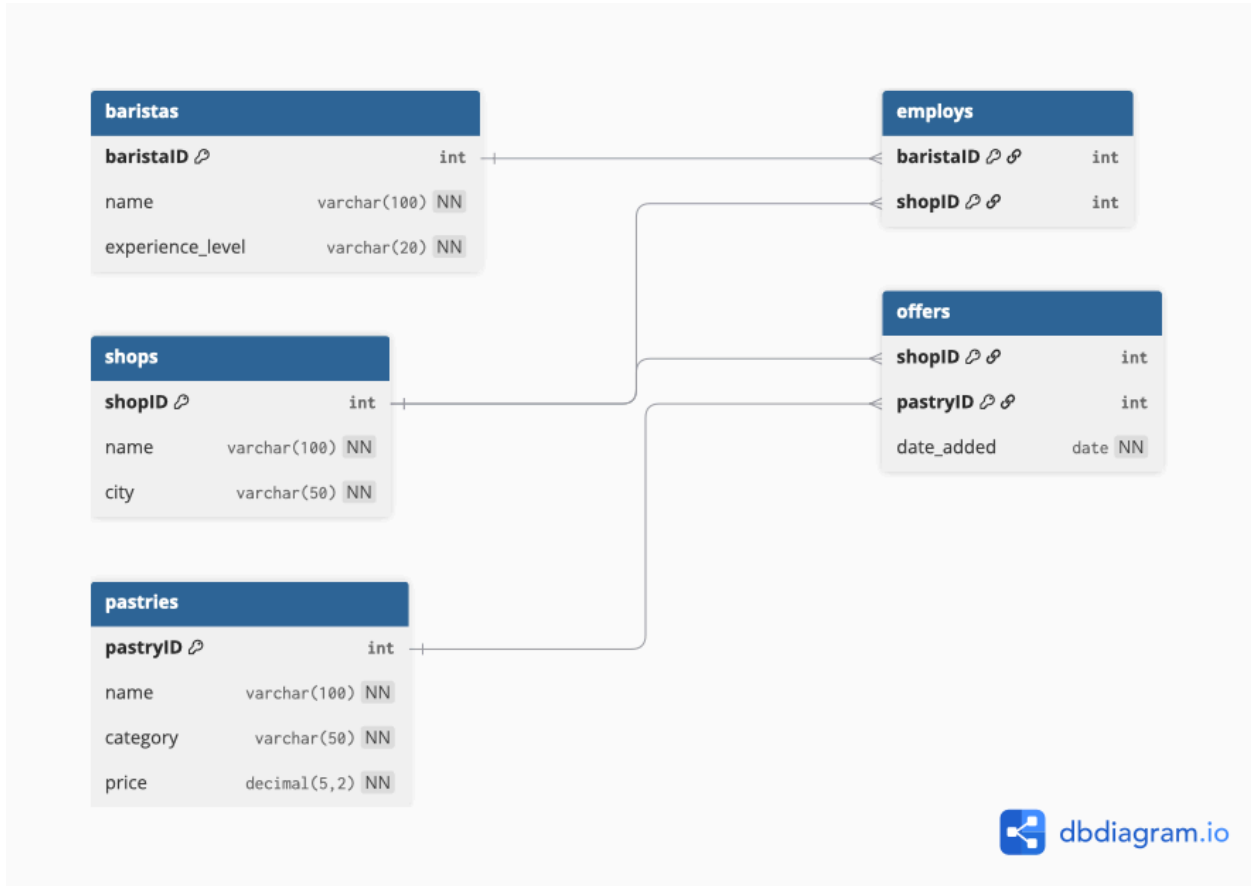


Najeeb Phillips
Assignment 2
Building a Small Relational Model in MySQL

Entity-Relationship Diagram



Problem 1: Average Price by Pastry Category
SELECT category, AVG(price) AS average_price
FROM pastries
GROUP BY category;

The screenshot shows the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The main window is divided into several panes:

- SCHEMAS:** A tree view on the left showing the database structure. The 'coffee_shop_db' database is selected, and its tables (baristas, employs, offers, pastries, shops) are listed.
- Query Editor:** The central pane where the SQL query is written. The query is:


```
1 • USE coffee_shop_db;
2
3 • SELECT category, AVG(price) AS average_price
4 FROM pastries
5 GROUP BY category;
6
7 • SELECT experience_level, COUNT(*) AS total_baristas
8 FROM baristas
9 GROUP BY experience_level;
10
11 • SELECT city, COUNT(*) AS total_shops
12 FROM shops
```
- Result Grid:** A table showing the results of the query. It has two columns: 'category' and 'average_price'. The results are:

category	average_price
Pastry	4.666667
Muffin	3.625000
Bakery	4.250000
Scone	3.850000
Breakfast	8.000000
Cake	3.875000
- Action Output:** A table at the bottom showing the execution of the query. It includes columns for Time, Action, Response, and Duration / Fetch Time. The query was executed at 20:47:23 and returned 6 rows.

On the right side of the interface, there is a 'Context Help' pane with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Problem 2: Total Baristas by Experience Level
 SELECT experience_level, COUNT(*) AS total_baristas
 FROM baristas
 GROUP BY experience_level;

The screenshot shows the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The main window is divided into several panes:

- Schemas:** A tree view on the left showing the database structure. The 'coffee_shop_db' database is selected, showing tables like baristas, employs, offers, pastries, shops, and views.
- Query Editor:** The central pane where the SQL query is written. The query is:


```

      5 GROUP BY category;
      6
      7 SELECT experience_level, COUNT(*) AS total_baristas
      8 FROM baristas
      9 GROUP BY experience_level;
      10
      11 SELECT city, COUNT(*) AS total_shops
      12 FROM shops
      13 GROUP BY city;
      14
      15 SELECT category, MAX(price) AS maximum_price
      16 FROM pastries
      
```
- Result Grid:** A table showing the results of the query. It has two columns: 'experience_level' and 'total_baristas'. The results are:

experience_level	total_baristas
Junior	2
Senior	2
Mid-Level	2
- Action Output:** A table at the bottom showing the execution of the query. It includes columns for Time, Action, Response, and Duration / Fetch Time. The query was executed at 20:53:25 and returned 3 row(s).

On the right side of the interface, there is a 'Context Help' pane with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Problem 3: Total Shops in Each City

```
SELECT city, COUNT(*) AS total_shops
FROM shops
GROUP BY city;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'coffee_shop_db' selected. The main editor contains three SQL queries. The first query is highlighted and its results are shown in the 'Result Grid' below. The 'Result Grid' shows a table with two columns: 'city' and 'total_shops'. The data rows are: Seattle (1), Portland (1), Austin (1), Chicago (1), Denver (1), and Boston (1). The bottom panel shows the 'Action Output' with a table of query execution details.

Query 1

```
10
11 • SELECT city, COUNT(*) AS total_shops
12 FROM shops
13 GROUP BY city;
14
15 • SELECT category, MAX(price) AS maximum_price
16 FROM pastries
17 GROUP BY category;
18
19 • SELECT shopID, COUNT(*) AS pastries_added
20 FROM offers
21 GROUP BY shopID;
```

Result Grid

city	total_shops
Seattle	1
Portland	1
Austin	1
Chicago	1
Denver	1
Boston	1

Result 62

Action Output

Time	Action	Response	Duration / Fetch Time
20:47:23	SELECT category, AVG(price) AS average_price FROM pastries GROUP BY category...	6 row(s) returned	0.0015 sec / 0.00000...
20:53:25	SELECT experience_level, COUNT(*) AS total_baristas FROM baristas GROUP BY ex...	3 row(s) returned	0.0010 sec / 0.00001...
20:56:06	SELECT city, COUNT(*) AS total_shops FROM shops GROUP BY city LIMIT 0, 1000	6 row(s) returned	0.0014 sec / 0.00001...

Problem 4: Maximum Pastry Price by Category

```
SELECT category, MAX(price) AS maximum_price
FROM pastries
GROUP BY category;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'coffee_shop_db' selected. The main editor shows a SQL query: `SELECT category, MAX(price) AS maximum_price FROM pastries GROUP BY category;`. The 'Result Grid' at the bottom displays the results of the query, showing 63 rows. The first few rows are:

category	maximum_price
Pastry	5.50
Muffin	3.75
Bakery	4.25
Scone	3.85
Breakfast	8.50
Cake	4.00

The bottom panel shows the 'Action Output' with a table of query execution details:

Time	Action	Response	Duration / Fetch Time
20:53:25	SELECT experience_level, COUNT(*) AS total_baristas FROM baristas GROUP BY ex...	3 row(s) returned	0.0010 sec / 0.00001...
20:56:06	SELECT city, COUNT(*) AS total_shops FROM shops GROUP BY city LIMIT 0, 1000	6 row(s) returned	0.0014 sec / 0.00001...
20:59:29	SELECT category, MAX(price) AS maximum_price FROM pastries GROUP BY category	6 row(s) returned	0.0010 sec / 0.00001...

Problem 5: Number of Pastries Offered by Each Shop
 SELECT shopID, COUNT(*) AS pastries_added
 FROM offers
 GROUP BY shopID;

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'coffee_shop_db' selected. The central editor shows a SQL query with line numbers 16 to 27. The query is:

```

16 FROM pastries
17 GROUP BY category;
18
19 • SELECT shopID, COUNT(*) AS pastries_added
20 FROM offers
21 GROUP BY shopID;
22
23 • SELECT p.name, p.category, p.price
24 FROM pastries AS p
25 WHERE p.price = (
26 SELECT MAX(p2.price)
27 FROM pastries AS p2
  
```

Below the query editor, the 'Result Grid' shows the results of the query. The grid has two columns: 'shopID' and 'pastries_add...'. The results are as follows:

shopID	pastries_add...
1	2
2	2
3	2
4	2
5	2
6	2

The bottom of the interface shows the 'Action Output' tab with a table of query execution details:

	Time	Action	Response	Duration / Fetch Time
✓	20:56:06	SELECT city, COUNT(*) AS total_shops FROM shops GROUP BY city LIMIT 0, 1000	6 row(s) returned	0.0014 sec / 0.00001...
✓	20:59:29	SELECT category, MAX(price) AS maximum_price FROM pastries GROUP BY category	6 row(s) returned	0.0010 sec / 0.00001...
✓	21:02:19	SELECT shopID, COUNT(*) AS pastries_added FROM offers GROUP BY shopID LIMIT 0, 1000	6 row(s) returned	0.0013 sec / 0.00000...

The status bar at the bottom indicates 'Query Completed'.

Problem 6: Highest-Priced Pastry in Each Category

```
SELECT p.name, p.category, p.price
FROM pastries AS p
WHERE p.price = (
    SELECT MAX(p2.price)
    FROM pastries AS p2
    WHERE p2.category = p.category
);
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' panel with 'coffee_shop_db' selected. The main editor shows a SQL query with syntax errors. The 'Result Grid' at the bottom displays the results of the query, showing a list of pastries with their names, categories, and prices. The 'Action Output' panel at the bottom shows the execution log, including the query text and the response from the database.

SQL Query:

```
23 SELECT p.name, p.category, p.price
24 FROM pastries AS p
25 WHERE p.price = (
26     SELECT MAX(p2.price)
27     FROM pastries AS p2
28     WHERE p2.category = p.category
29 );
30
31 SELECT DISTINCT o.shopID
32 FROM offers AS o
33 WHERE o.pastryID IN (
34     SELECT p.pastryID
```

Result Grid:

name	category	price
Almond Croissant	Pastry	5.50
Blueberry Muffin	Muffin	3.75
Cinnamon Roll	Bakery	4.25
Chocolate Scone	Scone	3.85
Blueberry Scone	Scone	3.85
Breakfast Burrito	Breakfast	8.50
Lemon Loaf	Cake	4.00

Action Output:

Time	Action	Response	Duration / Fetch Time
21:02:19	SELECT shopID, COUNT(*) AS pastries_added FROM offers GROUP BY shopID LIMIT 10	6 row(s) returned	0.0013 sec / 0.00000...
21:05:19	WHERE p.price = (SELECT MAX(p2.price) FROM pastries AS p2 WHERE p2.c...	Error Code: 1064. You have an error in your SQL syn...	0.0028 sec
21:06:20	SELECT p.name, p.category, p.price FROM pastries AS p WHERE p.price = (SELE...	7 row(s) returned	0.0012 sec / 0.00001...

Problem 7: Shops Offering Pastries Above the Overall Average Price

```
SELECT DISTINCT o.shopID
FROM offers AS o
WHERE o.pastryID IN (
    SELECT p.pastryID
    FROM pastries AS p
    WHERE p.price > (
        SELECT AVG(price)
        FROM pastries
    )
);
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'coffee_shop_db' selected. The main editor shows a SQL query with line numbers 37 to 48. The query is as follows:

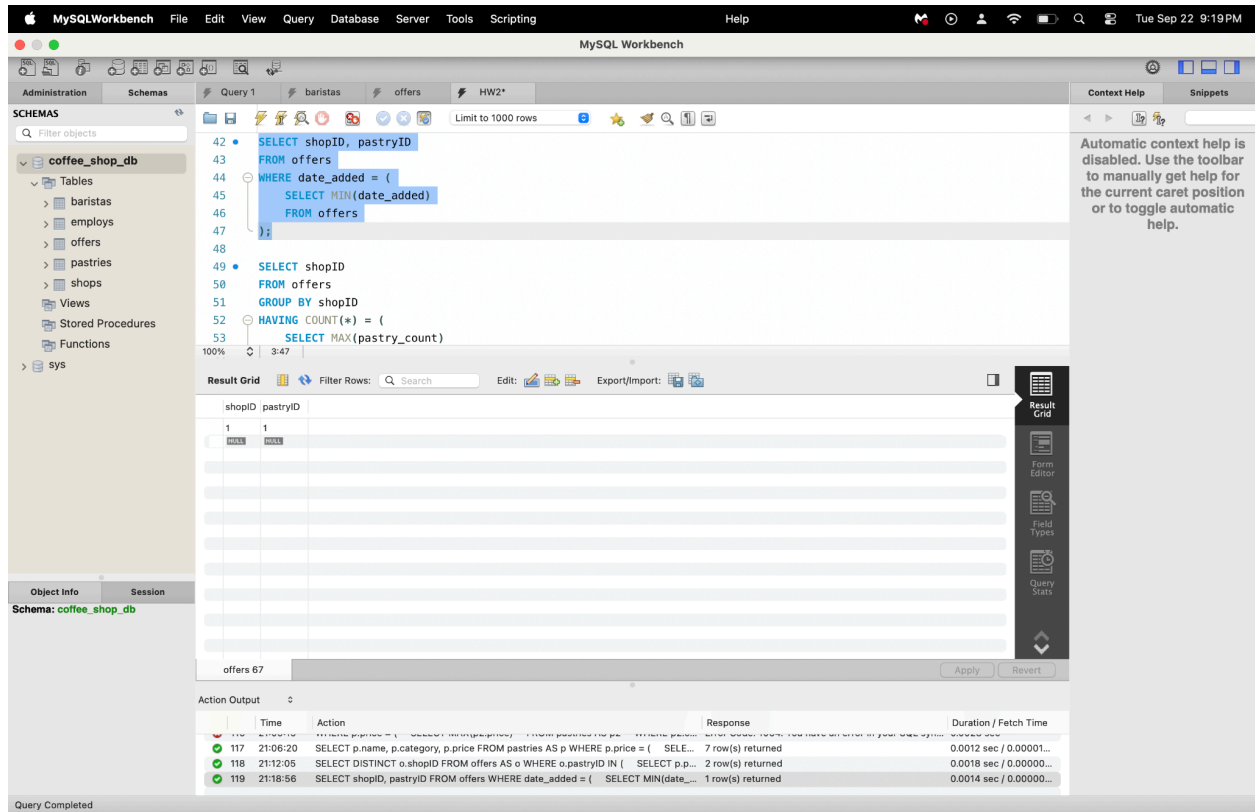
```
37 SELECT AVG(price)
38 FROM pastries
39 )
40 );
41
42 • SELECT shopID, pastryID
43 FROM offers
44 WHERE date_added = (
45     SELECT MIN(date_added)
46     FROM offers
47 );
48
```

The 'Result Grid' at the bottom shows the results of the query. It has two columns: 'shopID' and 'pastryID'. The first row shows '1' and '5'. The second row is empty. The status bar at the bottom indicates 'offers 66' and 'Read Only'.

The 'Action Output' pane at the bottom shows the execution log:

Time	Action	Response	Duration / Fetch Time
21:05:19	SELECT shopID, pastryID FROM offers WHERE date_added = (SELECT MIN(date_added) FROM offers)	7 row(s) returned	0.0012 sec / 0.00001...
21:06:20	SELECT p.name, p.category, p.price FROM pastries AS p WHERE p.price > (SELECT AVG(price) FROM pastries)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECT AVG(price) FROM pastries' at line 1	0.0028 sec
21:12:05	SELECT DISTINCT o.shopID FROM offers AS o WHERE o.pastryID IN (SELECT p.pastryID FROM pastries AS p WHERE p.price > (SELECT AVG(price) FROM pastries))	2 row(s) returned	0.0018 sec / 0.00000...

Query Completed

$$);$$


Problem 9: Shops Offering the Highest Number of Pastries

```
SELECT shopID
FROM offers
GROUP BY shopID
HAVING COUNT(*) = (
    SELECT MAX(pastry_count)
    FROM (
        SELECT COUNT(*) AS pastry_count
        FROM offers
        GROUP BY shopID
    ) AS shop_counts
);
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'coffee_shop_db' selected. The main editor shows a SQL query (lines 51-61) that finds shops with the highest number of pastries. The 'Result Grid' at the bottom shows the query results, with 'shopID' as the column and 6 rows of data. The 'Action Output' pane at the bottom shows the execution log for three queries, including the one being executed.

MySQL Workbench

Query 1

```
51 GROUP BY shopID
52 HAVING COUNT(*) = (
53     SELECT MAX(pastry_count)
54     FROM (
55         SELECT COUNT(*) AS pastry_count
56         FROM offers
57         GROUP BY shopID
58     ) AS shop_counts
59 );
60
61
```

Result Grid

shopID
1
2
3
4
5
6

offers 68

Action Output

Time	Action	Response	Duration / Fetch Time
21:12:05	SELECT DISTINCT o.shopID FROM offers AS o WHERE o.pastryID IN (SELECT p.p.p...	2 row(s) returned	0.0018 sec / 0.00000...
21:18:56	SELECT shopID, pastryID FROM offers WHERE date_added = (SELECT MIN(date_...	1 row(s) returned	0.0014 sec / 0.00000...
21:21:27	SELECT shopID FROM offers GROUP BY shopID HAVING COUNT(*) = (SELECT M...	6 row(s) returned	0.0012 sec / 0.00001...

Query Completed

Problem 10: Baristas Working at Shops in Seattle

```
SELECT name
FROM baristas
WHERE baristaID IN (
  SELECT baristaID
  FROM employs
  WHERE shopID IN (
    SELECT shopID
    FROM shops
    WHERE city = 'Seattle'
  )
);
```

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL query:

```
SELECT baristaID
FROM employs
WHERE shopID IN (
  SELECT shopID
  FROM shops
  WHERE city = 'Seattle'
);
```

The query results are displayed in the Result Grid, showing the following data:

name
Emma...
Liam...

The Action Output pane at the bottom shows the execution details of the query:

Time	Action	Response	Duration / Fetch Time
21:29:11	SELECT shopID, pastryID FROM offers WHERE date_added = (SELECT MIN(date_...	1 row(s) returned	0.00037 sec / 0.0000...
21:29:11	SELECT shopID FROM offers GROUP BY shopID HAVING COUNT(*) = (SELECT M...	6 row(s) returned	0.00037 sec / 0.0000...
21:29:11	SELECT name FROM baristas WHERE baristaID IN (SELECT baristaID FROM em...	2 row(s) returned	0.00047 sec / 0.0000...